

To determine the modulus of rigidity of a wire by Oscillating rod.

APPARATUS:

An oscillating cylindrical rod, a rigid support, meter rod, vernier callipers, screw guage, stop watch and wire of suitable length and radius. and Spring balance.

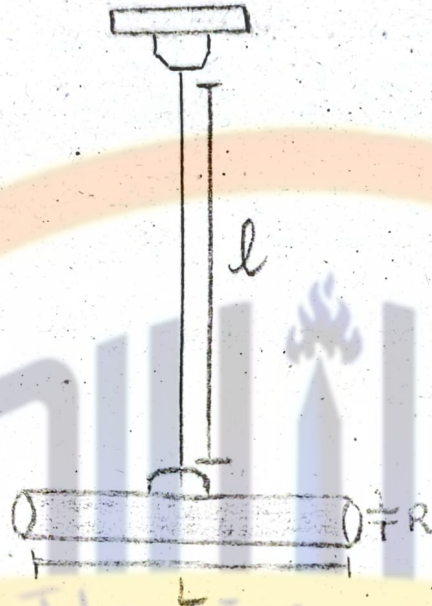
PROCEDURE:

- Suspend an oscillating metallic rod horizontally from one end of the wire and the other end is fixed in a rigid support.
- Slightly move the rod and note the position where you released the rod. After oscillating when it comes to the same position, 1 vibration will be completed.
- Start the stopwatch and find out the time ' t ' for 15 vibrations thrice and take their mean.
- Calculate the time period ' T ' and ' T^2 '.
- Measure the length ' L ' of the oscillating rod and length ' l ' of the suspension wire by a metre rod.

EXPERIMENT

To determine the modulus of rigidity of a wire by oscillating rod.

DIAGRAM:



OBSERVATIONS and CALCULATIONS:

1) For wire:

Length of the wire = $l = 93\text{cm}$

Least count of screw gauge = 0.001cm

Zero Error of screw gauge = Null

Diameter of the wire = $d = 0.083\text{cm}$

Radius of wire = $r = d/2 = 0.0415\text{cm}$

Experiment Name:

- Measure mass ' M ' of the rod by Spring balance
- Measure the diameter ' D ' of rod and diameter ' d ' of the wire with the vernier callipers and screw guage respectively.
- Calculate the radius " $R = D/2$ " of rod and radius " $r = d/2$ " of suspension wire.
- Calculate the moment of inertia ' I ' by using formula:

$$I = M \left(\frac{L^2}{12} + \frac{R^2}{4} \right)$$

- With the record of observations, Modulus of rigidity of wire is calculated by the relation

$$\eta = \frac{8\pi L}{r^4 T^2} \times I$$

For Rod:

Length of oscillating rod = 30.5 cm

L.C of Vernier Callipers = 0.01 cm

Zero error = Null

Diameter of the rod = $D = 2.20$ cm

Radius of the rod = $D/2 = 1.10$ cm

$$\text{Moment of Inertia} = I = M \left(\frac{L^2}{12} + \frac{R^2}{4} \right) \\ = 73153.93 \text{ gcm}^2$$

No. of Obs.	Time for 15 vibrations			Mean Time	Time Period $T = t/15$ sec	T^2
	1 sec	2 sec	3 sec			
1	133.2	132	129	131.4	8.76	76.73

Calculation:

$$\text{Modulus of rigidity } n = \frac{8\pi I}{r^4 T^2}$$

$$= \frac{8(3.14)(93)(73153.93)}{(0.0415)(76.73)}$$

$$= 7.559 \times 10^{11} \text{ dynes cm}^{-2}$$

Result:

Modulus of Rigidity n of wire is

$$7.559 \times 10^{11} \text{ dynes cm}^{-2}$$